Performance Benchmark of Elliptic Curve Cryptography on Microblaze Processor

A modern public-key encryption technique

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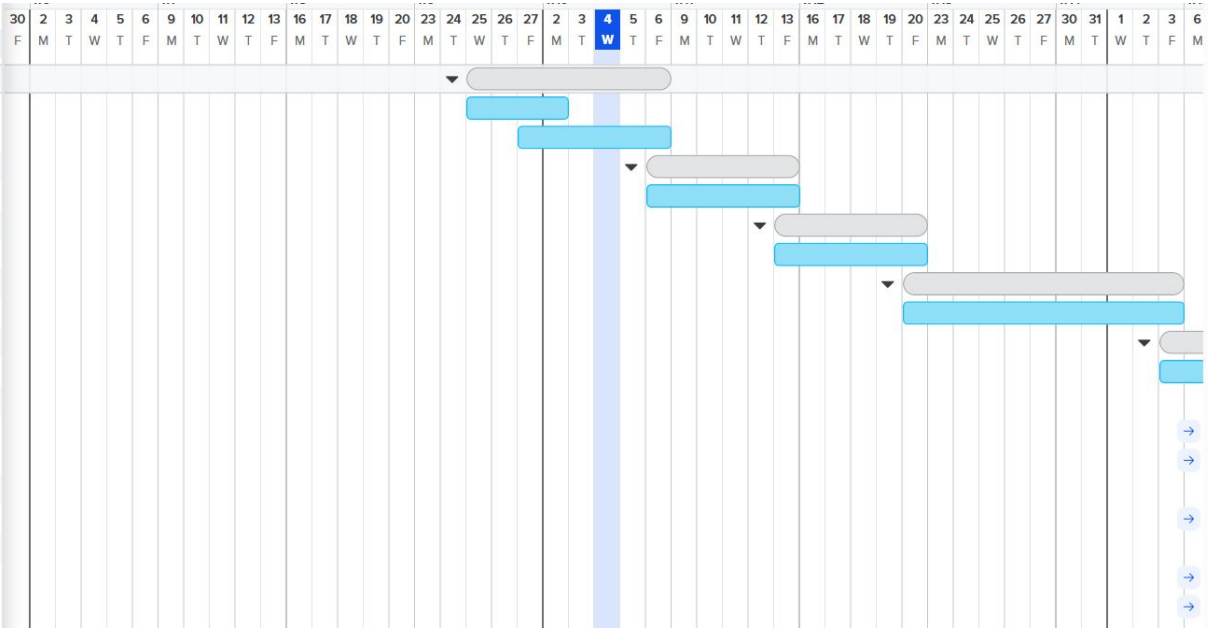
Project Overview

Goal: To evaluate the efficiency of ECC (secp256r1, secp192r1 and secp160r1) on Microblaze processor using Nexys A7-100T to determine its viability.

Why: To simulate real embedded systems and determine whether strong encryption like ECC can be efficiently used on resource-constrained systems such as embedded processors and IoT devices.

Benchmarks:

- **Latency and Throughput:** Cycles required for Key Generation, ECDSA Signing, and Verification.
- **Comparison:** A baseline test against the key sizes of secp256r1, secp160r1, and secp192r1 prove ECC's efficiency in resource-constrained environments.

RISC-V Benchmarking Project	**0%**
Setup	**0%**
Install RISC-V toolchain	0%
Deploy vanilla VexRiscv core on ...	0%
Integration	**0%**
Research and port the 'bare-meta...	0%
Baseline	**0%**
Run initial ECC-256 tests and veri...	0%
Benchmarking	**0%**
Measure cycles/time for different ...	0%
Comparison	**0%**
Run RSA-2048/3072 benchmarks...	0%
Analysis	**0%**
Collect Vivado power/resource d...	0%
Generate performance graphs	0%
Demo Prep	**0%**
Program the Nexys A7 7-segment...	0%
Final	**0%**
Record the video demo for the 'D...	0%
Finalize the report	0%